

Mohamed Elsayed

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EDUCATION

University of Toronto

Bachelor of Applied Science in Electrical Engineering

Toronto, Ontario

Sep 2025 – May 2030

- **Relevant Coursework:** Data Structures & Algorithms (Python/C), Linear Algebra, Engineering Math & Computation, Calculus I/II

EXPERIENCE

Undergraduate Research Assistant

Ain Shams University

May 2026 – Present

Cairo, Egypt

- Evaluated quantum-dot photovoltaic architectures by investigating optical absorption, carrier recombination, and device-efficiency tradeoffs through simulation and literature analysis.
- Modeled optoelectronic device behavior using **Ansys Lumerical** to identify quantum-dot integration strategies for improved light harvesting and energy conversion.

Undergraduate Research Assistant

American University in Cairo (AUC)

July 2025 – Aug 2025

New Cairo, Egypt

- Automated precision optics experiments by developing a microcontroller-controlled translation-stage system.
- Improved experimental control capabilities by designing motor-control circuitry and integrating embedded software in **C++**.

Backend Intern

eVision

July 2024 – Aug 2024

New Cairo, Egypt

- Reduced manual data-processing effort by engineering a scalable file-parsing and ingestion pipeline in **Java**.
- Accelerated software delivery by collaborating with frontend, database, and QA teams in a cross-functional development environment.

Technical Lead

Al Mannar NGO

Aug 2023 – Mar 2026

Luxor, Egypt

- Centralized beneficiary and financial record management by developing backend systems and database infrastructure in **Python**.
- Improved operational reliability by building internal automation tools and structured data-management workflows.

Software Developer & Technical Animator

Self-Employed

Oct 2021 – July 2023

Remote

- Supported game experiences reaching **1.7B+ visits** and **250K+ concurrent users** by developing backend gameplay systems and anti-cheat infrastructure.
- Delivered custom animation and VFX solutions by building production pipelines with Lua, Python, Blender, and Moon Animator.

PROJECTS

Entanglement Noise Budgets for Gaussian Channel Cascades 🔗 — *Python*

Feb 2026

- Derived a closed-form recurrence describing entanglement evolution through cascaded phase-insensitive Gaussian channels.
- Obtained analytical limits on thermal-loss channels, quantum-limited amplifiers, span count, channel ordering, and stochastic entanglement outage.
- Implemented numerical validation and reproducible experiments comparing closed-form predictions against exact symplectic-spectrum calculations.

FPGA Timing Closure & Congestion Prediction System 🔗 — *Verilog, Python, PyTorch Geometric*

Apr 2026

- Generated FPGA timing and routing datasets by building an automated benchmark flow using Yosys and nextpnr-ice40.
- Predicted timing closure risk, critical-path delay, and routing congestion by training **PyTorch Geometric GNNs** on graph representations of synthesized netlists.
- Improved evaluation reliability by implementing leakage-safe dataset splitting, multi-seed benchmarking, and end-to-end inference workflows.

Structural Beam Simulation 🔗 — *Python, MATLAB*

Nov 2025

- Simulated structural response for arbitrary beam geometries by implementing numerical analysis of loads, supports, and boundary conditions.
- Automated stress and deflection calculations by computing centroids, moments of inertia, and bending profiles for custom cross-sections.
- Validated simulation accuracy by comparing results against analytical solutions and physical beam-testing measurements.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Verilog, Java, JavaScript, TypeScript, SQL, MATLAB, HTML/CSS, Lua

Hardware & Electronics: Lumerical, FPGA, Microcontrollers, Motor Control, Circuit Design, PCB Assembly, Sensors

Developer Tools: Git, CUDA, RoCM, WSL, Codex, Claude Code

Libraries: PyTorch, PyTorch Geometric, NumPy, Pandas, Matplotlib